

Your Institution
Your Department

Friday 31st October, 2025

Scholarship Committee / Admissions Committee

Dear Members of the Committee,

I am delighted to provide a strong recommendation for [Candidate Name], whom I have supervised for approximately seven months on a challenging research project in maritime radar-camera sensor fusion. I can speak to their exceptional technical capabilities, research initiative, and genuine potential for continued growth as a researcher and engineer.

Technical Excellence

[Candidate] demonstrates exceptional technical breadth that is genuinely unusual at the Master's level. They have developed competence across multiple sophisticated domains simultaneously—radar signal processing, computer vision, deep learning, and systems engineering—while maintaining depth in each area. Their foundational knowledge of marine radar physics, CFAR detection algorithms, and neural network architectures is strong and well-applied.

Beyond theoretical understanding, [Candidate] has shown a remarkable ability to build real systems. They implemented a working Constant False Alarm Rate detector on actual radar data, designed a multi-sensor calibration framework, and coordinated the hardware integration of radar and optical systems. Rather than limiting themselves to simulation or theory, they conducted field data collection and worked with real hardware, which demonstrates maturity and practical engineering sense.

Their ability to articulate complex technical problems is clear and compelling. The project's core motivation—using radar's all-weather detection capabilities to compensate for camera limitations in maritime environments—shows sophisticated understanding of how to combine complementary sensor strengths for practical advantage. Their literature review synthesized over 131 references across multiple fields into coherent problem framing, demonstrating exceptional research organization skills.

Research Initiative and Communication

[Candidate] has consistently demonstrated professional research practices. They delivered monthly progress reports on schedule, maintained regular team meetings, and have begun mentoring undergraduate researchers—clear evidence of emerging leadership capacity. Their presentations are well-organized and communicate complex technical ideas effectively to audiences with varying technical backgrounds. This communication ability is increasingly rare and valuable in technical fields.

Areas for Growth

I want to be transparent about areas where [Candidate] would benefit from continued development, as this represents honest feedback that will help identify the most productive path forward.

[Candidate] excels at conceptual development and theoretical motivation. Their technical breadth and problem identification skills are genuine strengths. However, their research would be strengthened through increased emphasis on completing integrated systems and quantitative validation. During the project, individual components (CFAR detection, CNN models, multi-gain methodology) developed well in isolation, but end-to-end integration remained incomplete. This suggests that disciplined prioritization between exploration and completion would enhance their effectiveness. Successful research often requires resisting intellectually attractive tangents in favor of completing foundational work first.

Additionally, [Candidate] would strengthen their research methodology by emphasizing quantitative validation earlier in the development process. Claims about performance improvements are sometimes stated theoretically—“should reduce labeling time” or “expected to improve detection”—before corresponding measurements are gathered. In machine learning and signal processing, measurement and comparison against clear baselines are essential; approaches cannot be considered validated without quantitative rigor and statistical analysis. This is an area where structured feedback and mentorship would yield rapid improvement, as it reflects approach rather than capability.

When encountering technical obstacles, [Candidate] sometimes pivots to alternative approaches without fully documenting diagnostic steps or recovery attempts. Strengthening systematic problem-solving documentation—hypothesis, test, result, analysis—would increase confidence in their troubleshooting methodology. This too is readily addressable through experience and is not a fundamental limitation.

Recommendation

I strongly recommend [Candidate] for scholarship support and graduate programs. Their technical knowledge is strong and genuine, and their growth areas—focus, quantitative rigor, systematic documentation—are directly addressable through experience, mentorship, and feedback.

[Candidate] would be most successful in environments that emphasize project completion and integration as key milestones alongside technical exploration, that require quantitative validation before considering work complete, and that provide structured feedback on prioritization and scope management. They would also thrive in collaborative research settings where their communication skills and ability to understand complex multi-domain problems can be leveraged effectively.

For a scholarship supporting deep technical knowledge, broad learning ability, and strong research initiative, I recommend [Candidate] without reservation. They have demonstrated the foundation needed for success and possess the intellectual maturity to continue developing

the discipline and rigor that characterize exceptional researchers.

They represent a strong investment in a talented researcher with clear potential for continued development.

Sincerely,

Your Name